

MAT1002 Lecture 22,

Outline

- Line integral of $\vec{F}$ along C
- Flow, circulation and flux
- Conservative fields
 - ↳ Gradient fields
 - ↳ The loop property

Line Integrals of $\vec{F}$ along C

Computation

For a space curve C given by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, ($a \leq t \leq b$)

$$d\vec{r} = \langle dx, dy, dz \rangle.$$

If $\vec{F} = \langle M, N, P \rangle$, ← $M(x, y, z)$ then

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_C \vec{F} \cdot \vec{r}'(t) dt = \int_C M dx + N dy + P dz \\ &= \int_C M dx + \int_C N dy + \int_C P dz. \end{aligned}$$

For computation, we may change all variables into the parameter t ;

e.g., $\int_C M dx = \int_a^b M(x(t), y(t), z(t)) x'(t) dt$, etc.

Note that $\int_C M dx + N dy + P dz = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$.

Example

Find the work done by the force field $\mathbf{F} = \langle x, y, z \rangle$ in moving a particle along the curve C parametrized by

$$\mathbf{r}(t) = \langle \cos(\pi t), t^2, \sin(\pi t) \rangle, \quad t \in [0, 1].$$

Ans: $\frac{1}{2}$.

Flows and Circulations

Whenever $\vec{F}$ is a velocity field of a flowing fluid, the integral $\int_C \vec{F} \cdot d\vec{r}$ is often called a flow integral. Whenever

C is a closed curve (i.e., $\vec{r}(a) = \vec{r}(b)$), the flow is often called a **circulation**.

- $\int_C \vec{F} \cdot \vec{T} ds$: "line integral of $\vec{F}$ along C "; "flow integral"
- When C is closed, we may write $\oint_C \vec{F} \cdot \vec{T} ds$ instead of $\int_C \vec{F} \cdot \vec{T} ds$
- Note that the orientation (direction) of C matters for flows and circulations, since reversing C changes the direction of $\vec{T}$.

↙ We may write $\oint_C$ for $\int$ to emphasize the orientation of C

Simple Curves

A curve is said to be **simple** if it does not cross itself.

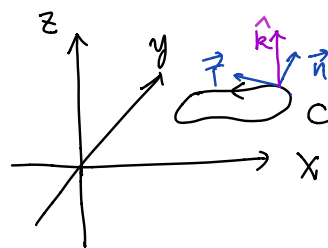
Flux (for Vector Fields in $\mathbb{R}^2$)

Def: If C is a simple closed curve and $\vec{F} = \langle M(x,y), N(x,y) \rangle$ is a vector field in $\mathbb{R}^2$, and $\vec{n}$ is the outward normal to C , then the **(outward) flux of $\vec{F}$ across C** is the integral

$$\int_C \vec{F} \cdot \vec{n} ds.$$

If $\vec{F}$ is the velocity field of some flowing fluid, then the (outward) flux measures the tendency of the fluid flowing out of C .

Using the Right-hand rule, one can see that
 if C is counterclockwise, then $\vec{n} = \vec{T} \times \hat{k}$, and
 if C is clockwise, then $\vec{n} = \hat{k} \times \vec{T}$.



Assuming that C is oriented counterclockwise,

$$\begin{aligned}\vec{n} &= \vec{T} \times \vec{k} = \frac{d\vec{r}}{ds} \times \vec{k} = \left\langle \frac{dx}{ds}, \frac{dy}{ds}, 0 \right\rangle \times \langle 0, 0, 1 \rangle \\ &= \frac{dy}{ds} \vec{i} - \frac{dx}{ds} \vec{j}\end{aligned}$$

Hence, $\vec{n} ds = dy \vec{i} - dx \vec{j}$, and

$$\boxed{\int_C \vec{F} \cdot \vec{n} ds = \oint_C \vec{F} \cdot \vec{n} ds = \oint_C (M dy - N dx) .}$$

e.g. Find the flux of $\vec{F} = \langle x-y, x \rangle$ across the circle C
 $x^2 + y^2 = 1$.

Sol: C : $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, counterclockwise,
 does not repeat except $\vec{r}(0) = \vec{r}(2\pi)$.

$$dy = \cos t dt, \quad dx = -\sin t dt$$

$$\Rightarrow \text{flux} = \oint_C \vec{F} \cdot \vec{n} ds = \oint_C M dy - N dx$$

$$= \int_0^{2\pi} (\cos t - \sin t) \cos t dt + \cos t \sin t dt = \int_0^{2\pi} \cos^2 t dt$$

$$= \int_0^{2\pi} \frac{1 + \cos(2t)}{2} dt = \pi.$$

Conservative Fields

A conservative force is a force with the property that the total work done in moving a particle between two points is independent of the path taken.

Definition

Let $\mathbf{F}$ be a vector field defined on an open region D . Then $\mathbf{F}$ is said to be **conservative on D** if the following condition holds: for any two points A and B in D , if C_1 and C_2 are piecewise smooth curves in D from A to B , then

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}.$$

Any such line integral, which depends only on the initial and terminal points (but not the path itself), is said to be **path independent**.

Consider the vector field $\mathbf{F}(x, y) := \langle -y, x \rangle$. One can check that

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} \neq \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$$

if C_1 is the line segment from $(0, 0)$ to $(1, 1)$ and C_2 is the curve given by

$$\langle t, t^2 \rangle, \quad t \in [0, 1].$$

Although both C_1 and C_2 are smooth curves from $(0, 0)$ to $(1, 1)$, the line integrals of $\mathbf{F}$ along them are different.

Hence, the field $\vec{F} := -y\vec{i} + x\vec{j}$ is not conservative.

Q: Which fields are conservative? What properties do they have?

Theorem (Fundamental Theorem of Line Integrals) (FTLI)

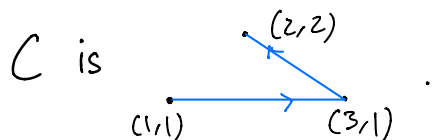
Let C be a piecewise smooth curve from A to B , parametrized by $\mathbf{r}(t)$, with $t \in [a, b]$. Let f be a real-valued function such that ∇f is continuous on a region D containing C . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

Proof: Shown in class (also in the book). □

As an immediate consequence, any continuous gradient field is conservative.

E.g. Find the work done by force $\vec{F} = \langle x, y \rangle$ along C , where



Sol: $\vec{F} = \nabla f$, where $f(x,y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$. FTLI

$$\begin{aligned} \text{Work} &= \int_C \vec{F} \cdot d\vec{r} = \int_{(1,1)}^{(2,2)} \nabla f \cdot d\vec{r} = f(2,2) - f(1,1) \\ &= \frac{1}{2}(4+4) - \frac{1}{2}(1+1) = 4 - 1 = 3. \end{aligned}$$

Definition

If $\mathbf{F} = \nabla f$, then f is called a **potential function** for $\mathbf{F}$.

Q: Are there other conservative fields other than gradient fields?

A: No (assuming continuity).

Definition

A region D is said to be connected if any two points in D are connected by a curve lying entirely in D .

Theorem

Let $\mathbf{F}$ be a vector field whose components are continuous on an open connected region D . Then $\mathbf{F}$ is **conservative** on D if and only if $\mathbf{F}$ is a **gradient field** (that is, $\mathbf{F} = \nabla f$ for some f defined on D).

Will prove this for fields in $\mathbb{R}^2$. A proof for $\mathbb{R}^3$ is in the book (and is similar).

Proof: · “ $\Leftarrow$ ”: This follows from the Fundamental Theorem of Line Integrals.

· “ $\Rightarrow$ ”: Assume that $\vec{F}$ is conservative.

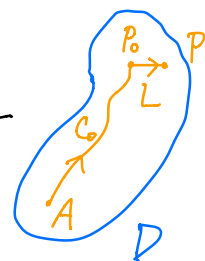
· Fix $A := (a,b) \in D$. Define $f: D \rightarrow \mathbb{R}$ by $f(x,y) := \int_{(a,b)}^{(x,y)} \vec{F} \cdot d\vec{r}$
exists by path independence

· Consider $P := (x,y)$. (what is f_x at (x,y) ?)

· Fix $P_0 := (x_0, y_0)$ in D (say $x_0 < x$).

· Let C_0 be a smooth curve from A to P_0 , and let L be the line segment from P_0 to P .

· Now $f(x,y) = \int_A^P \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot d\vec{r}$
curve from A to P



$$= \underbrace{\int_{C_0} \vec{F} \cdot d\vec{r}}_K + \int_L \vec{F} \cdot d\vec{r} \quad L: \vec{r}(t) = \langle t, y \rangle, \quad x_0 \leq t \leq x$$

$$= K + \int_{x_0}^x \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = K + \int_{x_0}^x M(t, y) dt$$

$$\vec{r}'(t) = \langle 1, 0 \rangle$$

- Consider moving the variable point P along the x -axis by a bit:

$$\frac{\partial}{\partial x} f(x, y) = 0 + \frac{\partial}{\partial x} \int_{x_0}^x M(t, y) dt \stackrel{\text{By FTC}}{=} M(x, y).$$

- Similarly, $\frac{\partial}{\partial y} f(x, y) = N(x, y)$.

- Hence, $\nabla f = \vec{F}$; $\vec{F}$ is a gradient field. □

The Loop Property

- A given parametrization $\mathbf{r}(t)$, $a \leq t \leq b$, determines an **orientation** (or direction) of a curve C , with the positive direction corresponding to increasing values of t .
- In general, if C is a curve with a given orientation, we use $-C$ to denote **the same curve with the opposite orientation**. Then

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}.$$

Theorem

The following statements are equivalent for any vector field $\mathbf{F}$.

- (a) The field $\mathbf{F}$ is conservative on D .
- (b) For every closed curve C in D , $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.

Means: (a) $\Leftrightarrow$ (b); (a) if and only if (b).

Proof: Shown in class. □

Example

The vector field defined by

$$\mathbf{F}(x, y) := \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$$

is not conservative on $\mathbb{R}^2 \setminus \{(0, 0)\}$, since its line integral along the unit circle is not equal to 0.